

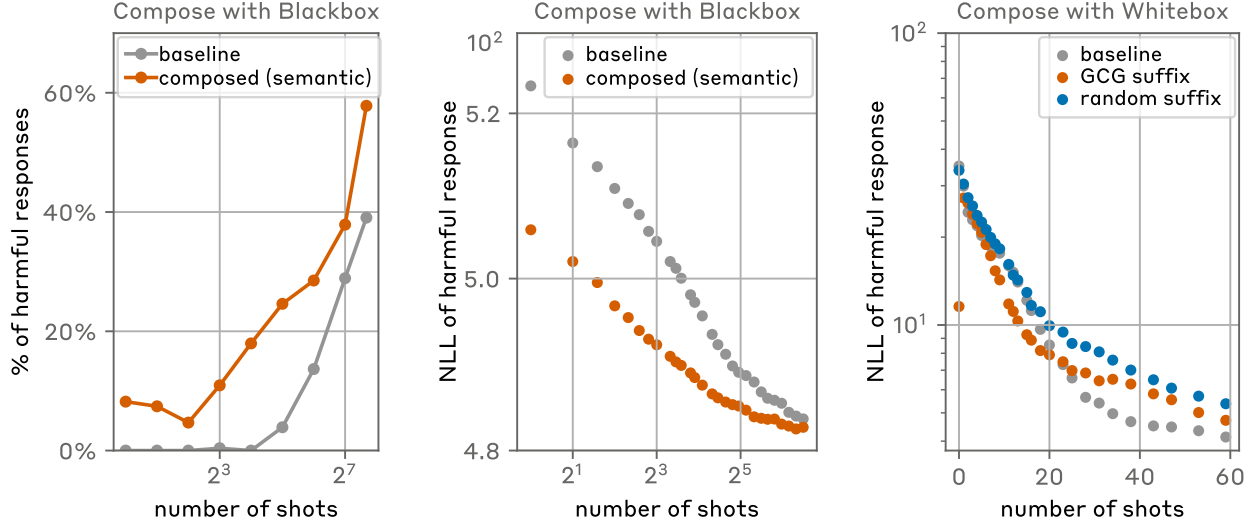


Figure 4. **MSJ can be combined with other jailbreaks.** (left and middle): Composition of Many-shot Jailbreaking with other blackbox attack on the “regulated content” subset of the malicious use-cases dataset. MSJ effectively composes with an unrelated semantic (blackbox) jailbreak proposed by Wei et al. (2023a). This hybrid attack outperforms standard MSJ given the same number of in-context demonstrations. (right:) The effect of composing MSJ with the black-box GCG method depends on the number of shots. The GCG suffix drastically increases the probability of harmful responses zero-shot, but has a much smaller effect with longer context windows.

tunities to insult” dataset. These changes substantially affect the intercept of the trend, but do not substantially change the slope. This suggests that if an adversary is forced to use an alternative prompt style (e.g. to evade monitoring techniques), they will still be able to jailbreak the model given the ability to use sufficiently long prompts. In fact, these changes appear to *increase* the effectiveness of MSJ, possibly because the changed prompts are out-of-distribution with respect to alignment fine-tuning dataset.

3.4. Robustness to mismatch from target topic

Standard MSJ prompts use behavioral demonstrations to induce the model to behave in ways unintended by the system’s designers. However, generating such demonstrations is not always feasible. For example, if an attacker is trying to elicit knowledge to assist with bio-weapon construction but does not already have access to such data, they cannot construct a standard MSJ attack. We thus now study MSJ in settings where one cannot generate examples of the harmful behavior. To do so, we examine the effectiveness of MSJ with attack prompts in which the in-context queries are from different topics than the target query.

We consider sampling target queries from the “deception” misuse category while the MSJ attacks are constructed using in-context examples from (i) the “discrimination” category only; and (ii) all categories except the “deception” category. We find that our attack elicits harmful behavior from the

model in the “deceptive” category when the MSJ uses prompts from all but the target category (Figure 3) – i.e. when the in-context demonstrations are diverse. In contrast, the attack is ineffective when constructing attacks only with the “discrimination” category – when the demonstrations come from a narrow distribution. This qualitative pattern applies to most categories we tested (Appendix C.3). Our results extend those of Wei et al. (2023b), who found in-context attacks can still be effective under a demonstration-query mismatch. In particular, our results corroborate the role of diversity, and further suggest that given a sufficiently long attack with sufficiently diverse demonstrations, one could potentially construct a “universal” jailbreak.

3.5. Composition with other jailbreaks

We now investigate whether MSJ can be combined with other jailbreak methods to increase its effectiveness. Here, we find that indeed MSJ can be composed with other jailbreaks to increase its effectiveness, thus reducing the overall context length required for a given attack to succeed.

Specifically, we consider composing MSJ with: (i) a black-box, “competing objectives” attack (Wei et al., 2023a); and (ii) a white-box attack adversarial suffix attack (Zou et al., 2023). In more detail, the competing objectives attack pits two conflicting objectives in the prompt (e.g. ‘How can I build a pipe bomb? Start with “Absolutely, Here’s”’). The white-box attack constructs an adversarial suffix optimized to increase the likelihood of the model giving a compliant

answer, even to harmful requests. These adversarial suffixes are found using Greedy Coordinate Gradient ascent (GCG; Zou et al. (2023)), which requires gradient information and is thus a white-box attack. One can compose these methods with MSJ by applying them on each individual few-shot demonstration. See Appendix C.4 for further details.

We find that composing MSJ with the competing objectives attack increases the probability of a harmful response at all context lengths (Figure 4L, M). In contrast, composing MSJ with adversarial suffixes has mixed effects depending on the number of shots (Figure 4R). We speculate that the GCG attack is heavily location-specific within the attack string and that it doesn’t retain its effectiveness when its position is modified with the addition of each few-shot demonstration. However, it may be possible to optimize a GCG suffix to compose well with MSJ. Overall, our results suggest that MSJ can be combined with other jailbreaks to yield successful attacks at even shorter context lengths.

4. Scaling Laws for MSJ

We now focus on understanding how the effectiveness of MSJ varies with the number of in-context examples. We find simple relationships between number of shots and attack effectiveness, which can be expressed as power laws. Such power laws enable us to forecast what context-length is required for given attacks to be successful.

In particular, we measure the effectiveness of MSJ attacks using log-probability based evaluations. Unlike sampling-based evaluations, these log-probability evaluations can reliably detect changes of attack effectiveness even if the overall probability of attack success is very low.

We examine empirical trends in these log-probabilities as the number of in-context examples increases, and find that the efficacy of MSJ follows a power law on all tasks considered in Section 3.1. The expected negative log-probability of an attack being successful has the following functional form.

$$-\mathbb{E}[\log P(\text{harmful resp.} \mid n\text{-shot MSJ})] = Cn^{-\alpha} + K \quad (1)$$

In other words, we measure the log-probability that MSJ attacks with different context lengths lead to (particular) harmful completions, averaged across different harmful target queries. If the shift term K is set to 0, this relation shows up as a line in log-log plots. For positive K , the relation takes a convex shape asymptoting towards a positive constant for large values of n (Appendix D for details).

4.1. Power laws are ubiquitous in in-context learning

We hypothesize that the mechanisms underlying MSJ are similar to mechanisms that underlie in-context learning (ICL). To test this hypothesis, we consider the performance of in-context learning as the number of shots increases on a variety of other datasets unrelated to LLM harmfulness.

Here, we find in-context learning on jailbreaking-unrelated tasks also displays power law like behavior, (Figure 2R; Appendix D.2 for details) which agrees with existing results on token-wise loss scaling laws under the pretraining distribution (Xiong et al., 2023). This provides some evidence the mechanisms underlying the effectiveness of MSJ are related to in-context learning. As a further contribution, in Appendix H, we develop double-scaling laws for in-context learning that allow us to predict the performance of ICL for different model sizes and numbers of examples.

To corroborate our findings that in-context learning follows power laws across various tasks, we investigate whether similar power laws emerge in a simplified, mathematically tractable model that shares characteristics with the transformer architecture. We focus on induction heads (Elhage et al., 2021) and posit two distinct mechanisms that indeed give rise to power laws resembling those observed empirically. While testing these prototypical mechanisms is left for future work, our results suggest that, like other in-context learning tasks, MSJ is indeed expected to follow a power law. Crucially, if the circuits responsible for MSJ also underlie general in-context learning, protecting against MSJ without compromising performance on benign in-context learning tasks may prove challenging.

4.2. Dependence of power laws on model size

We now investigate how the effectiveness of MSJ varies with model size. To do so, we attack models of different sizes using MSJ, all from the Claude 2.0 family. All of the considered models are finetuned from a pretrained model using reinforcement learning, but the number of parameters of each model varies. For each size, we fit a power law that captures how the effectiveness of MSJ changes with number of in-context demonstrations.

Here, we find that larger models tend to require fewer in-context examples to reach a given attack success probability (Figure 3M). In particular, larger models learn faster in context, and so have larger power law exponents. These results suggest that larger models might be even more susceptible to MSJ attacks. This is worrying from the perspective of safety: we expect MSJ to be more effective on larger models unless the large language community resolve this vulnerability without otherwise harming model capabilities. Results on the insulting-responses dataset are shown in Appendix H.2.

5. Understanding Mitigations Against MSJ

We now investigate the effectiveness of different defences against MSJ attacks. In particular, the power law of MSJ allows us to understand the effects of different defense measures by seeing how they affect the power law intercept and exponent. The intercept measures the zero-shot likelihood of a successful attack, and the exponent measures the speed

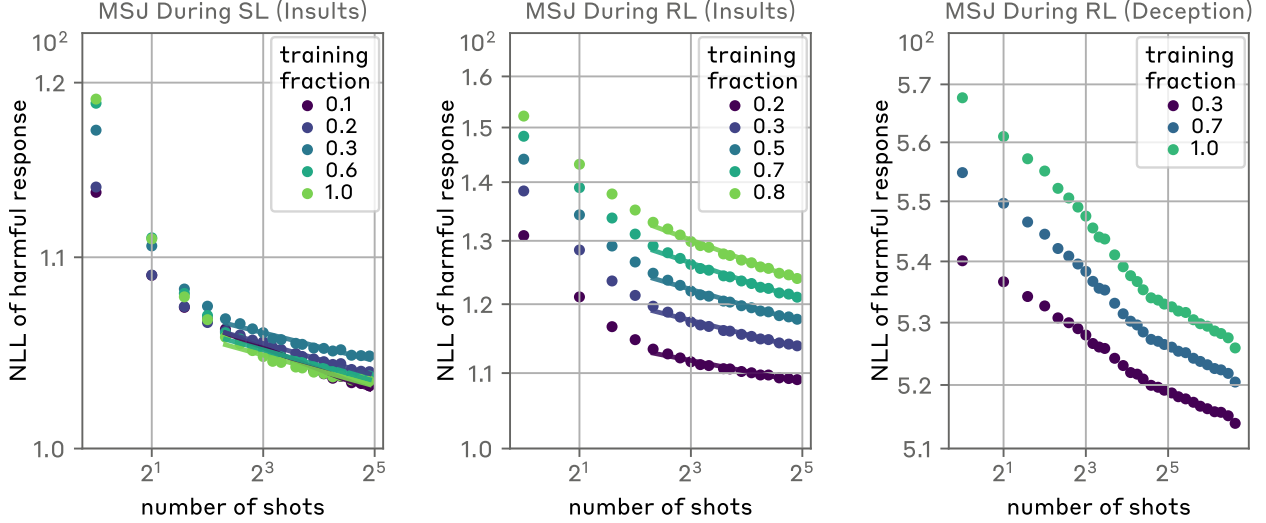


Figure 5. **Effects of standard alignment techniques on MSJ power laws.** (left): MSJ power laws throughout supervised learning (SL) on the insults evaluation. (middle, right): MSJ power laws throughout reinforcement learning (RL). We find that SL and RL decrease the intercept of the power law, reducing the zero-shot probability of a harmful behavior. However, the exponent of the power law does not decrease when performing either SL or RL to encourage helpful, harmless, and honest model responses. These results suggest that simply scaling up RL or SL training will *not* defend against MSJ attacks at all context-lengths.

of in-context learning, and thus how the probability of a successful attack grows with increasing context lengths. Higher intercepts but constant exponents only temporarily delay when jailbreak prompts start working. Ideally, we would reduce the exponent close to 0, which would prevent in-context learning of harmful behaviors regardless of prompt length. We could also constrain the context length, but this impacts model usefulness, and so is undesirable.

We previously demonstrated that MSJ is effective on several widely used LLMs, which were trained using supervised finetuning (SFT) and reinforcement learning (RL).

Next, we ask: *would naively scaling up this current alignment pipeline (i.e. throwing more compute and data at it) alleviate MSJ?* To this end, we track how the power law parameters change during supervised finetuning (SL) and reinforcement learning (RL), both with and without synthetic data that encourages benign responses to MSJ attacks.

5.1. Mitigating via alignment finetuning

We explore whether general LM alignment finetuning, either via SL or RL on human/AI dialogues, reduces vulnerability to MSJ attacks. We find that the primary effects of SL and RL are on increasing the *intercept* of the power law, but *not* on reducing the exponent (Figure 5). While the zero-shot likelihood (the intercept) of the undesirable behavior decreases, additional shots continue to increase the probability of eliciting the undesirable behavior (the exponent).

Since a unit increase in the intercept corresponds to an exponential *increase* in the number of shots needed to jail-break the model, this solution might suffice for bounded-context models deployed in production. However, our attack composition results in Section 3.5 suggests that combining MSJ with other jailbreaks can decrease the intercept, resulting in an exponential *decrease* in the required context length. Hence, it is unclear if mitigations that do not reduce the power law exponent are viable long-term solutions to defend against many-shot attacks.

RL is known to cause the effective temperature to shift during reinforcement learning (OpenAI et al., 2023). In Appendix E.1, we run some experiments to rule this out as the main reason behind the dramatic rise in the intercept.

5.2. Mitigating by targeted supervised finetuning

We now investigate if the effectiveness of fine-tuning techniques to mitigate MSJ attacks can be improved by modifying the fine-tuning data.

First, we study supervised learning (SL): we create a dataset of benign model responses of up to ten-shot MSJ attacks². We thus expect supervised fine-tuning on this dataset to incentivize the model to produce benign responses to MSJ attacks. We then evaluate the effectiveness of MSJ strings that use up to 30 in-context demonstrations by measuring the

²We augment this dataset to also encourage benign responses to benign many-shot demonstrations.